

Sammy Abdella

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EDUCATION

Columbia University in The City of New York

Expected 05/2027

B.S. Computer Science

SKILLS

Technical: Python, Git, Linux, REST, PostgreSQL, C++, Java, JavaScript, SQL, HTML/CSS, Ruby, OAuth2, JWT, Docker, Kubernetes

Languages: French, Arabic, English

EXPERIENCE

Web Developer Intern (Part-time) | CUIT | New York, NY

05/2025 – Present

- Migrated 43 Columbia departmental websites from Drupal 7 to Drupal 10, writing migration scripts, refactoring custom modules, building/refreshing site themes with HTML/CSS, running regression tests, improving average page load times by ~35%, and removing 100% of deprecated modules in production
- Integrated a Django, PostgreSQL, and RESTful API backend with a new Columbia admissions website, scaling reliably to 50,000+ applicants per cycle, cutting form submission errors by ~40%, and reducing manual data correction work by ~50%

Backend Software Developer Intern | Zitouna Bank | Remote

01/2024 – 05/2024

- Engineered RESTful APIs using Python/Django to cut response times by 30% and boost transaction throughput by 25%
- Improved PostgreSQL for real-time financial workloads by improving indexing and query plans; cut p95 latency ~35% and improved reliability under peak traffic
- Authored 50+ unit/integration tests for critical transaction endpoints and database logic; increased backend test coverage from 62% to 82% and reduced post-release defects by 25%

Research Intern | Columbia University AI Laboratory | New York, NY

01/2025 – 09/2025

- Led the development for gAyl, an AI tool for LGBTQ+ suicide prevention, designed six model-based safety benchmarks to generate treatment plan recommendations and achieved <15% false-negative rate on unsafe-response detection across 1,000+ high-risk prompts
- Trained LLM-based classifiers in PyTorch and Hugging Face Transformers to automate safety evaluation, reducing manual benchmarking time by ~95%
- Publication (co-author): <https://doi.org/10.7916/z6vc-d867>

Software Engineer Intern | Compyle.ai | San Francisco, California

05/2025 – 08/2025

- Added Sentry error monitoring and alerts for the summarization/Q&A pipeline; reduced time-to-detect production errors by ~60% and cut repeat-user failure reports by ~20%
- Implemented request throttling and caching for LLM calls using Redis; reduced average LLM API requests per user session by ~30% and lowered latency spikes during peak usage

Projects

AI-Powered Exam Tutor | Python, Django, PostgreSQL, Redis, Celery, Docker

09/2025 – 01/2026

- Built a web app that generates timed practice exams, stores attempts, and produces graded reports with explanations and targeted remediation suggestions
- Deployed with Columbia University's Learning Center to support 50+ students across Data Structures, Advanced Programming, and Multivariable Calculus exam prep